Global Superstore

Business Requirement Document and Data Understanding

Project: Power BI Analytics - Executive Sales and Profitability Dashboard

Prepared for: [Course / Program Name]

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1. Business Requirement Document

1.1 Background

Global Superstore is a multinational retailer selling Furniture, Office Supplies, and Technology products to Consumer, Corporate, and Home Office customers across multiple world markets. Leadership currently reviews performance through static spreadsheets and requires a self-service, interactive Power BI dashboard so that regional managers, finance, and operations can independently answer their own questions without waiting on a periodic report.

1.2 Objective

Deliver an interactive Power BI report that enables stakeholders to monitor sales, profitability, customer value, and shipping performance, filterable by year, market, segment, and product category.

1.3 Stakeholders

Stakeholder	Primary interest
Executive Leadership	Overall sales and profit health, year-over-year trend
Regional Sales Managers	Performance by Market and Region
Merchandising	Category, Sub-Category, and Product performance
Finance	Margins and discounting impact
Customer Success / Marketing	Customer value and segment behavior
Operations / Logistics	Shipping cost and delivery speed

1.4 Scope

In scope: a single-source data model built from one data table, an interactive multi-page Power BI report, filter controls, and a consistent visual theme.

Out of scope: forecasting and predictive modeling, real-time data refresh, row-level security, and external data blending. These items are flagged as potential enhancements for a future phase of this project.

1.5 Business Problem Statements

#	Theme	Problem statement
BRD-01	Sales Performance	Leadership needs one view of Total Sales, Profit, Orders, and Quantity, comparable against the prior year.
BRD-02	Sales Performance	Regional managers need to see which Markets and Regions drive the most revenue.
BRD-03	Sales Performance	Merchandising needs Sales and Profit broken down by Category and Sub-Category.

#	Theme	Problem statement
BRD-04	Profitability	Finance wants to quantify how Discount % relates to Profit.
BRD-05	Profitability	Finance wants the Top 10 products by Sales and Profit to prioritize promotion.
BRD-06	Profitability	Leadership wants Profit Margin % visible by Region and Category.
BRD-07	Customer	Customer Success wants Sales by Segment and the top customers by value.
BRD-08	Customer	Marketing wants unique customer counts and Average Order Value per Market.
BRD-09	Operations	Operations wants Shipping Cost as a % of Sales and average delivery delay by Ship Mode.
BRD-10	Operations	Operations wants Order Priority distribution to sanity-check service levels.

1.6 Success Criteria

- All 10 problem statements are answered by at least one visual in the report.
- The report responds to filter changes within approximately 2 seconds.
- Visual design is consistent: one theme, an aligned grid, and content that is readable at a glance.
- Every visual carries a clear title, updated dynamically where relevant.

2. Data Understanding

2.1 Grain

The table is structured at order-line grain: each row represents one product within one customer order. Row ID serves as the unique identifier for a row; Order ID is not unique per row, since an order containing multiple products generates multiple rows sharing the same Order ID. This distinction is foundational to the analysis, particularly when differentiating order-level metrics (such as number of orders) from line-item-level metrics (such as number of line items).

2.2 Data Dictionary

Column	Type	Notes
Row ID	Integer	Unique row-level identifier
Order ID	Text	Repeats across multi-product orders; not unique per row
Order Date / Ship Date	Date	Two separate date fields; analysis should specify which is being used
Ship Mode	Text	e.g. Standard Class, First Class, Second Class, Same Day
Customer ID / Customer Name	Text	One customer can place many orders
Segment	Text	Consumer, Corporate, Home Office
Postal Code	Decimal	Populated only for the United States portion of the data; nulls elsewhere are expected
City / State / Country / Region / Market	Text	Market is the top-level geography; Region sits beneath it
Product ID / Category / Sub-Category / Product Name	Text	Category to Sub-Category to Product Name forms a natural hierarchy
Sales / Profit / Shipping Cost	Decimal	Profit can be negative; this reflects real business outcomes, not an error
Quantity	Integer	Units per line
Discount	Decimal	Stored as a proportion, e.g. 0.2 represents 20%
Order Priority	Text	Low, Medium, High, Critical

2.3 Data Quality Observations and Assumptions

- Postal Code contains null values for the majority of rows. This is expected and not a data quality defect: postal codes apply only to the United States portion of the dataset and should not be imputed.
- Order ID does not represent the grain of the table. Any measure counting orders should use a distinct count of Order ID; any measure counting line items should use a row count. These are two different, equally valid metrics and should be labeled accordingly.

• Profit values can be negative. This reflects genuine business outcomes, such as heavily discounted or high-shipping-cost transactions, rather than a data error, and should not be excluded from analysis by default.

• Two separate date fields exist: Order Date and Ship Date. Analysis should explicitly state which date basis is being used, as this affects time-based aggregations and trend calculations.

Appendix A: Dataset Preview

The source table contains 51,290 rows and 24 columns. The schema and a single sample row are provided below for reference.

A.1 Column Schema

#	Column	Type	Non-null / total
0	Row ID	int64	51,290 / 51,290
1	Order ID	object	51,290 / 51,290
2	Order Date	datetime64[ns]	51,290 / 51,290
3	Ship Date	datetime64[ns]	51,290 / 51,290
4	Ship Mode	object	51,290 / 51,290
5	Customer ID	object	51,290 / 51,290
6	Customer Name	object	51,290 / 51,290
7	Segment	object	51,290 / 51,290
8	Postal Code	float64	9,994 / 51,290
9	City	object	51,290 / 51,290
10	State	object	51,290 / 51,290
11	Country	object	51,290 / 51,290
12	Region	object	51,290 / 51,290
13	Market	object	51,290 / 51,290
14	Product ID	object	51,290 / 51,290
15	Category	object	51,290 / 51,290
16	Sub-Category	object	51,290 / 51,290
17	Product Name	object	51,290 / 51,290
18	Sales	float64	51,290 / 51,290
19	Quantity	int64	51,290 / 51,290
20	Discount	float64	51,290 / 51,290
21	Profit	float64	51,290 / 51,290
22	Shipping Cost	float64	51,290 / 51,290
23	Order Priority	object	51,290 / 51,290

A.2 Sample Row

Column	Value
Row ID	40098
Order ID	CA-2014-AB10015140-41954
Order Date	2014-11-11
Ship Date	2014-11-13
Ship Mode	First Class
Customer ID	AB-100151402
Customer Name	Aaron Bergman
Segment	Consumer
Postal Code	73120.0
City	Oklahoma City
Product ID	TEC-PH-5816
Category	Technology
Sub-Category	Phones
Product Name	Samsung Convoy 3
Sales	221.980
Quantity	2
Discount	0.0
Profit	62.1544
Shipping Cost	40.77
Order Priority	High

Note: State, Country, Region, and Market values for this sample row were not included in the original data preview and are therefore omitted above rather than estimated.